



From Slab to Beam

Tributary Areas, Line Loads and a Two-Dimensional Subframe

Course	Computer (& AI) Methods for Structural Engineering (DISC 5203)
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Assigned plan	Floor Plan 1 (final digit 1)
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24 beam segments | net slab 180.128 m² | four OpenSees models | exact area and reaction equilibrium

Scope

This report extends the existing HW1 Git history. It reuses the HW1 slab and vertical-element results, transfers the same gravity load to beams with panel-specific 45-degree and mixed-support 60/30-degree boundaries, models the two marked beams as standalone continuous beams and one-floor subframes, and cross-checks subframe column forces against HW1 axial estimates.

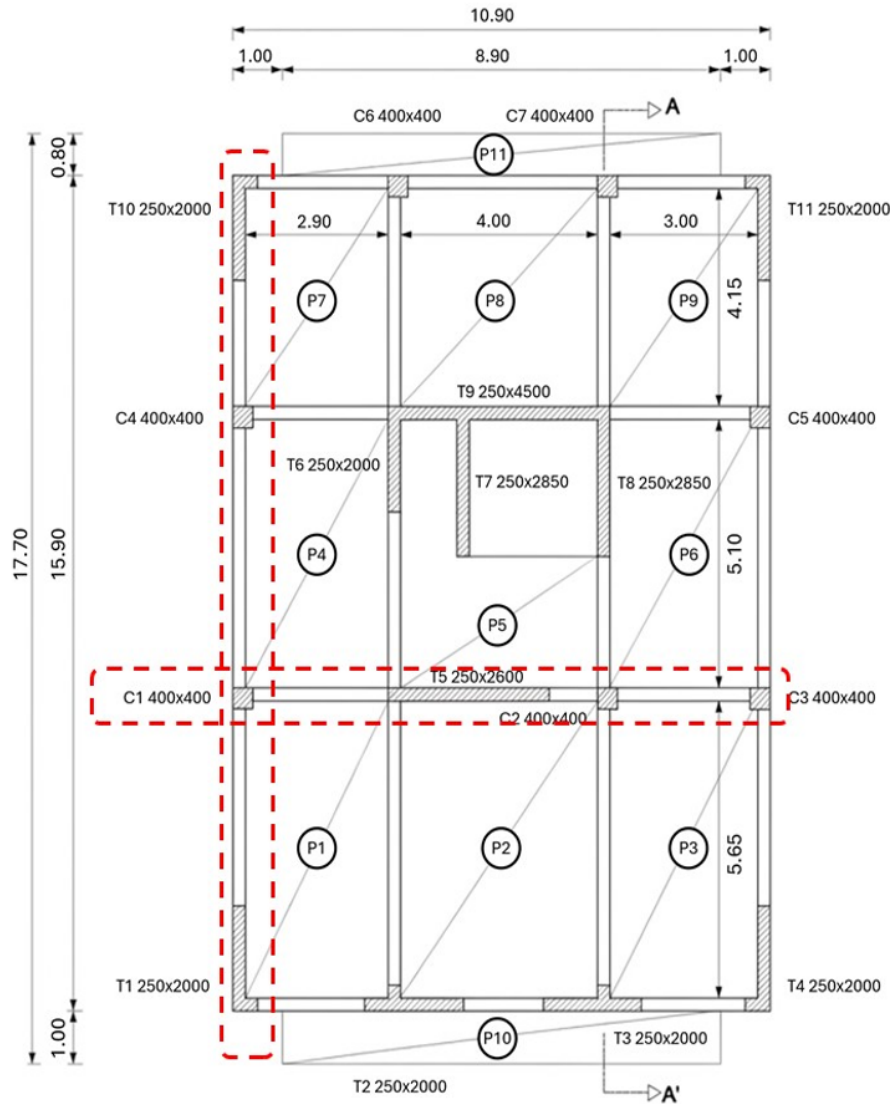


Figure 1. Assigned Floor Plan 1; red boxes mark Beam A and Beam B.

Dead slab load = $25 \times 0.150 = 3.75 \text{ kN/m}^2$; live load = 5.00 kN/m^2 . No finish load is printed. Beam self-weight is excluded from the requested slab-to-beam transfer. Unprinted beam size is explicitly assumed as $300 \times 600 \text{ mm}$; columns are $400 \times 400 \text{ mm}$, storey height 3.0 m and $E = 30.0\text{e}6 \text{ kN/m}^2$.

2. Panel 45-degree and mixed 60/30-degree construction

Exterior main-slab edges are discontinuous/simple and shared interior edges are continuous/fixed. Matched edge conditions divide at 45 degrees. At a mixed corner the boundary leaves the continuous edge at 60 degrees and the discontinuous edge at 30 degrees; this is implemented by weighted distance $d/\sqrt{3}$ versus d . Openings are removed before assignment. P10 and P11 are one-edge cantilevers and load only H0/H3.

Both total-load-equivalent and simply-supported midspan-moment-equivalent UDLs are reported. The stored non-uniform profile, not either tabulated UDL, is used for OpenSees local response.

HW2 Beam Tributary Areas - Panel 45-degree / Mixed 60-30 Construction

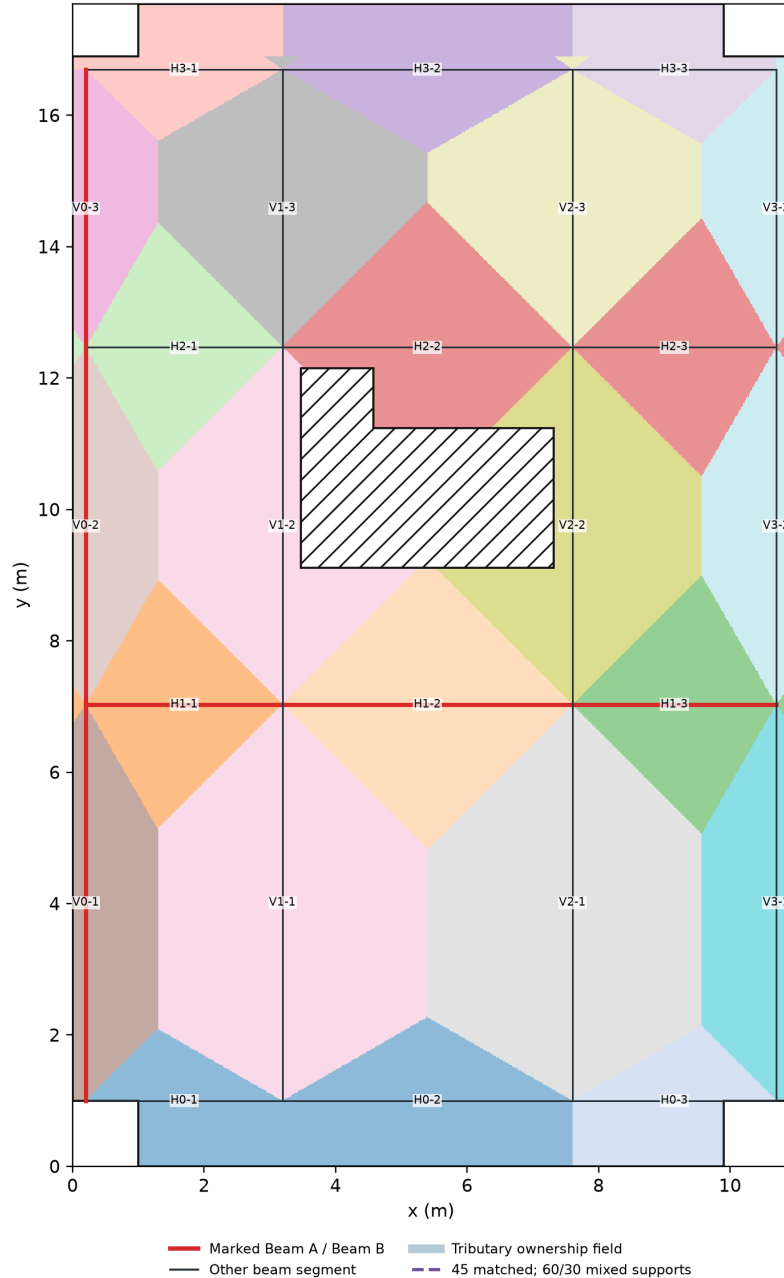


Figure 2. Beam ownership field: 45 degrees for matched supports and 60/30 degrees for mixed supports.

3. Beam census and equivalent line-load table

Load-eq preserves total load; M-eq preserves simply-supported midspan moment. The exact profile is retained for local shear and moment diagrams.

ID	Span	Panels	Class	Area	D load	L load	Tot load	Tot M-eq
H0-1	2.990	P1; P10 (cantilever balcony where overlapping)	two loaded sides locally (main slab + balcony)	3.846	4.824	6.432	11.255	13.712
H0-2	4.410	P2; P10 (cantilever balcony where overlapping)	two loaded sides locally (main slab + balcony)	7.205	6.127	8.169	14.295	16.146
H0-3	3.100	P3; P10 (cantilever balcony where overlapping)	two loaded sides locally (main slab + balcony)	4.069	4.922	6.563	11.485	13.995
H1-1	2.990	P1; P4	interior, slab on both sides	5.723	7.177	9.570	16.747	21.445
H1-2	4.410	P2; P5	interior, slab on both sides	9.707	8.254	11.005	19.259	25.585
H1-3	3.100	P3; P6	interior, slab on both sides	6.148	7.437	9.915	17.352	22.194
H2-1	2.990	P4; P7	interior, slab on both sides	5.723	7.177	9.570	16.747	21.445
H2-2	4.410	P5; P8	interior, slab on both sides	8.220	6.990	9.319	16.309	21.440
H2-3	3.100	P6; P9	interior, slab on both sides	6.148	7.437	9.915	17.352	22.194
H3-1	2.990	P7; P11 (cantilever balcony where overlapping)	two loaded sides locally (main slab + balcony)	4.000	5.016	6.688	11.705	13.904
H3-2	4.410	P8; P11 (cantilever balcony where overlapping)	two loaded sides locally (main slab + balcony)	7.150	6.080	8.107	14.187	16.137
H3-3	3.100	P9; P11 (cantilever balcony where overlapping)	two loaded sides locally (main slab + balcony)	4.184	5.062	6.749	11.811	14.182
V0-1	6.030	P1	perimeter, slab on one side	6.145	3.822	5.095	8.917	10.492
V0-2	5.440	P4	perimeter, slab on one side	4.918	3.390	4.520	7.911	9.777
V0-3	4.230	P7	perimeter, slab on one side	3.822	3.388	4.517	7.906	9.616
V1-1	6.030	P1; P2	interior, slab on both sides	18.090	11.250	15.000	26.250	32.089
V1-2	5.440	P4; P5	interior, slab on both sides	9.803	6.758	9.010	15.768	19.199
V1-3	4.230	P7; P8	interior, slab on both sides	10.687	9.474	12.632	22.106	28.069
V2-1	6.030	P2; P3	interior, slab on both sides	18.233	11.339	15.119	26.458	32.437
V2-2	5.440	P5; P6	interior, slab on both sides	10.352	7.136	9.515	16.651	19.956
V2-3	4.230	P8; P9	interior, slab on both sides	10.804	9.578	12.771	22.349	28.400
V3-1	6.030	P3	perimeter, slab on one side	6.264	3.895	5.194	9.089	10.744
V3-2	5.440	P6	perimeter, slab on one side	4.981	3.433	4.578	8.011	9.948
V3-3	4.230	P9	perimeter, slab on one side	3.908	3.465	4.620	8.084	9.757

Units: span in m; area in m²; line loads in kN/m.

4. Load profiles and worked example

For H1-2, the actual two-sided area is read from P2 and P5 after the core opening is removed using their classified 45/60/30 boundaries. $D = 3.75$ A and $L_{live} = 5.00$ A. Load-eq is total divided by span; M-eq is 8 times the simply-supported midspan moment divided by L_2 . The figure retains the polygonal distribution used in OpenSees.

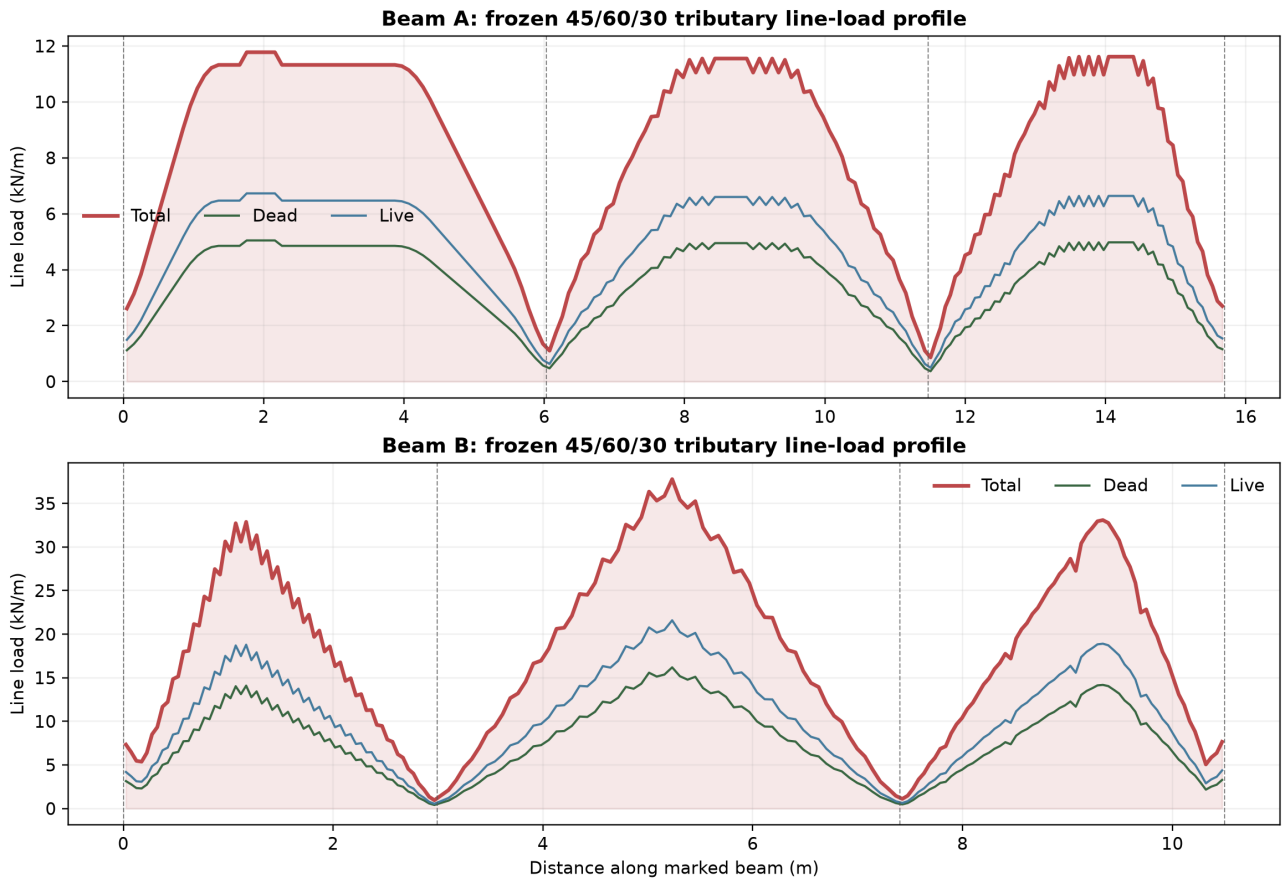


Figure 3. Frozen dead, live and total line-load profiles.

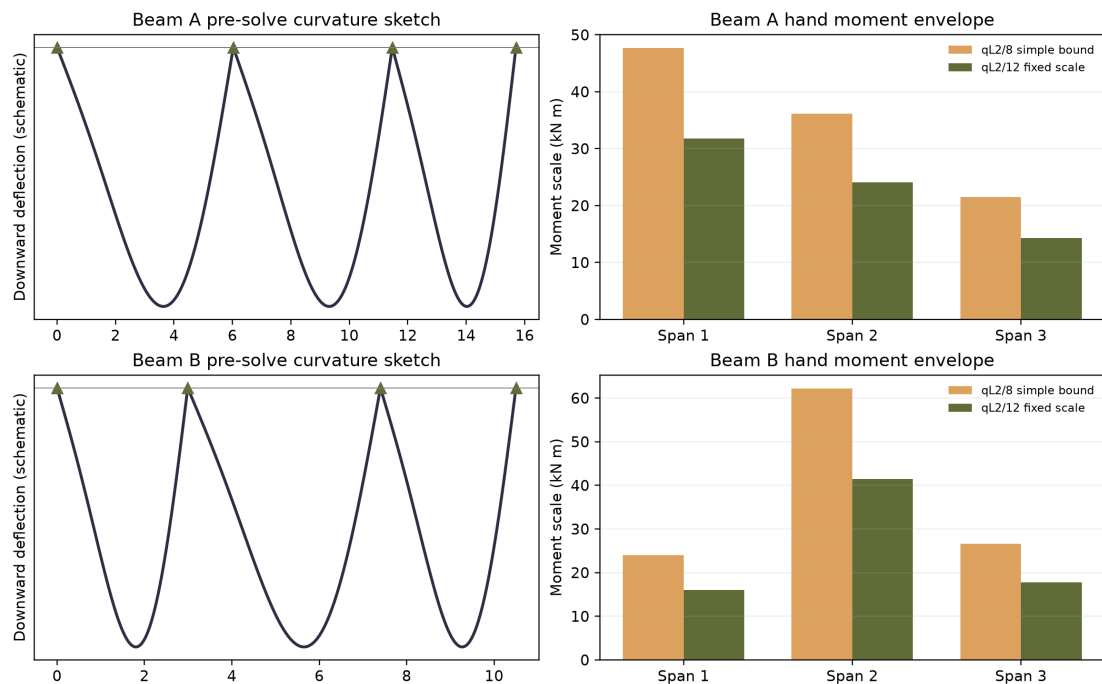


Figure 4. Pre-solve sketches and qL2 bounds.

5. OpenSees response diagrams

Standalone models retain vertical supports with free rotations. Subframes replace them with fixed-base 3 m columns sharing the beam nodes. The same frozen profile is applied in every comparison.

Beam A - Pure Model | applied 130.2 kN | equilibrium error 5.43e-09 kN

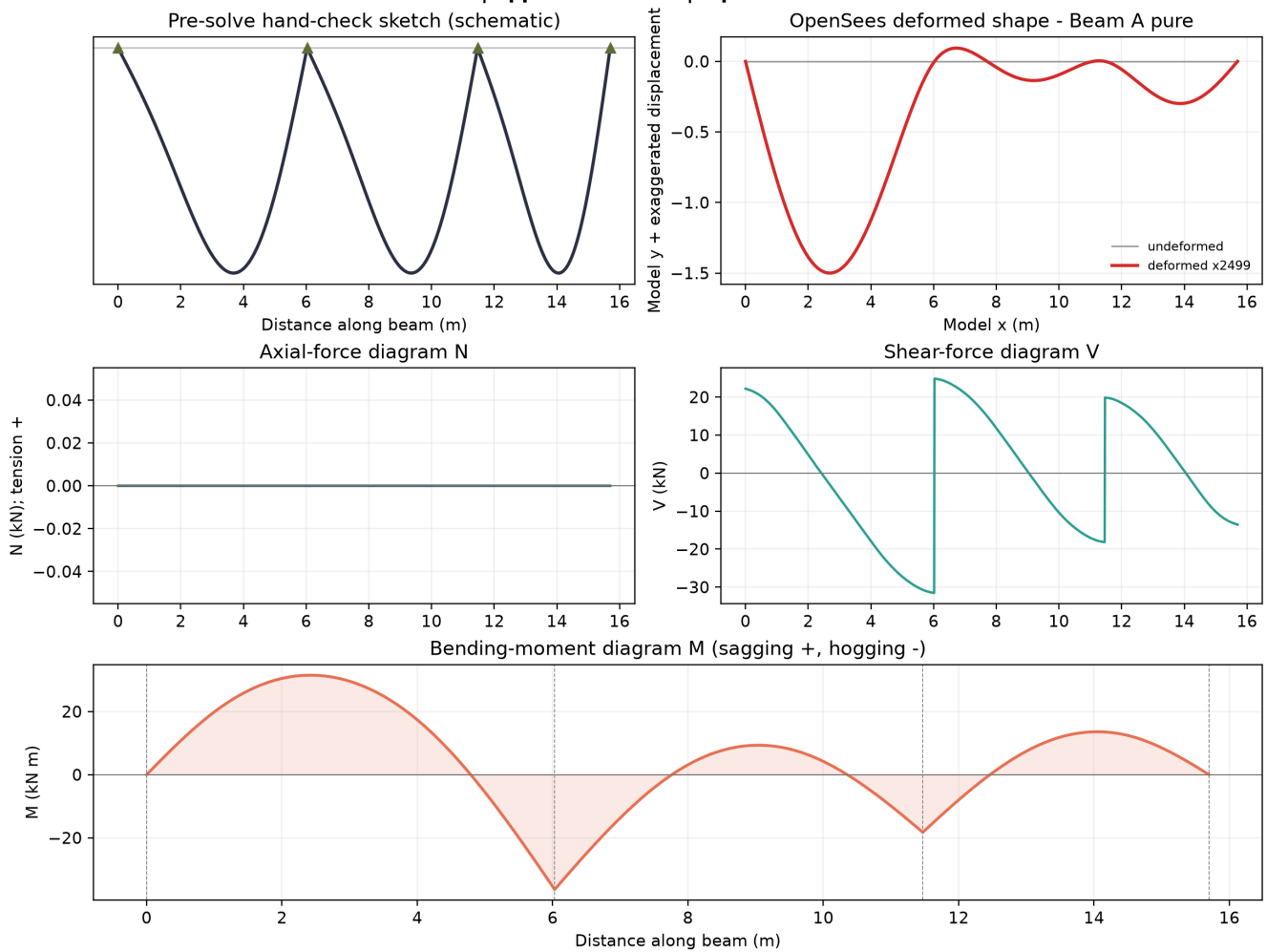


Figure. Beam A, pure model: pre-solve sketch, deformed shape and N/V/M diagrams.

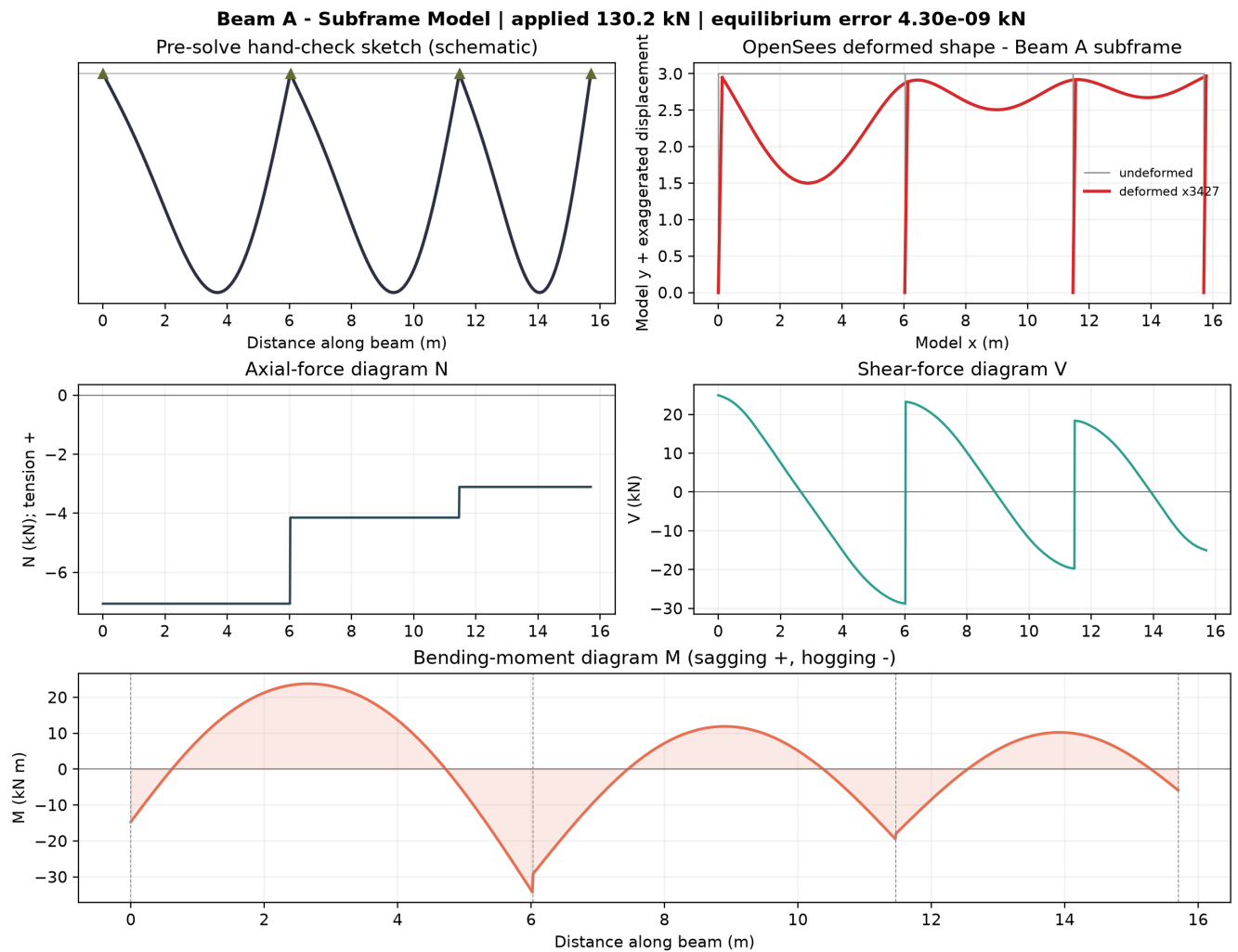


Figure. Beam A, subframe model: pre-solve sketch, deformed shape and N/V/M diagrams.

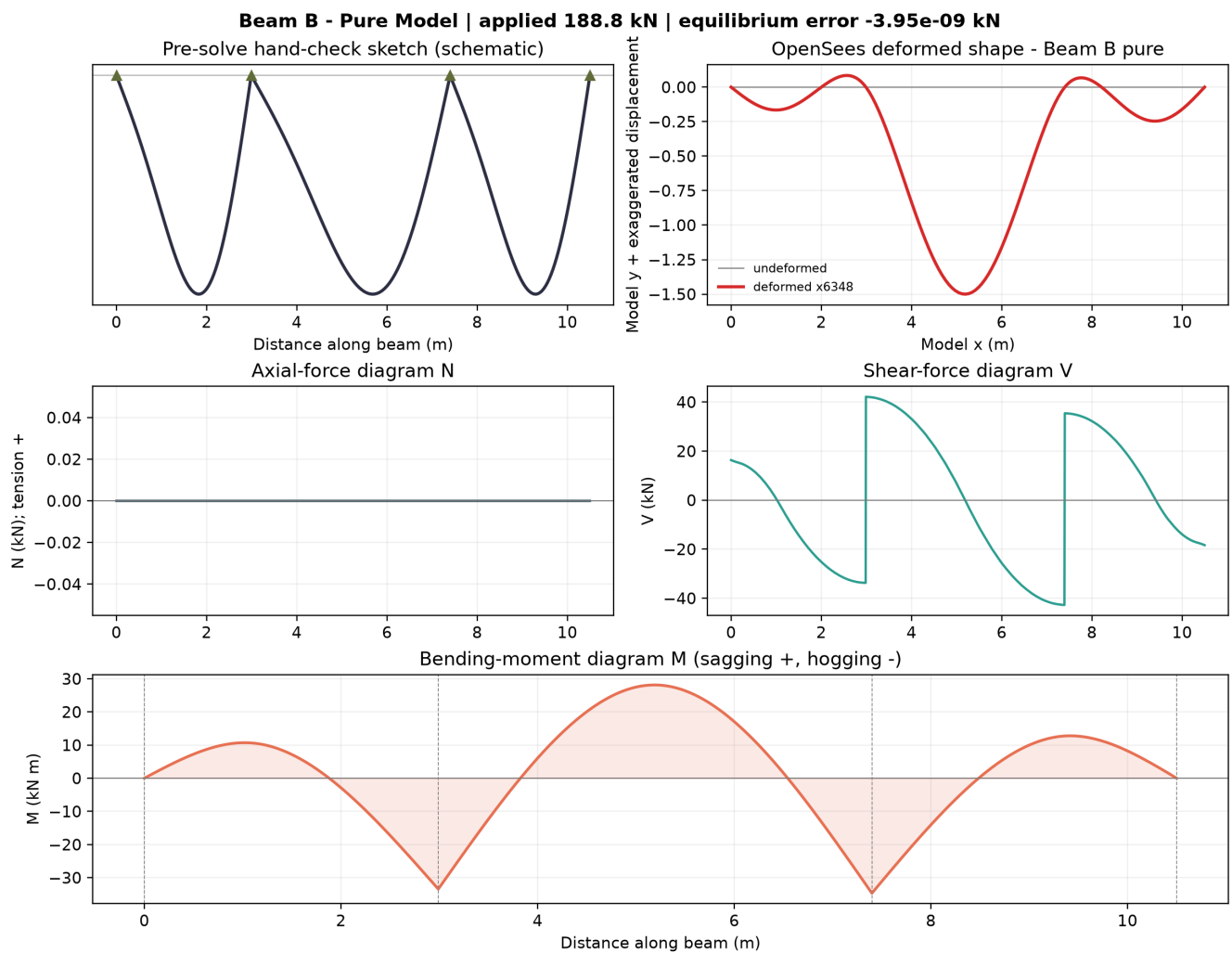


Figure. Beam B, pure model: pre-solve sketch, deformed shape and N/V/M diagrams.

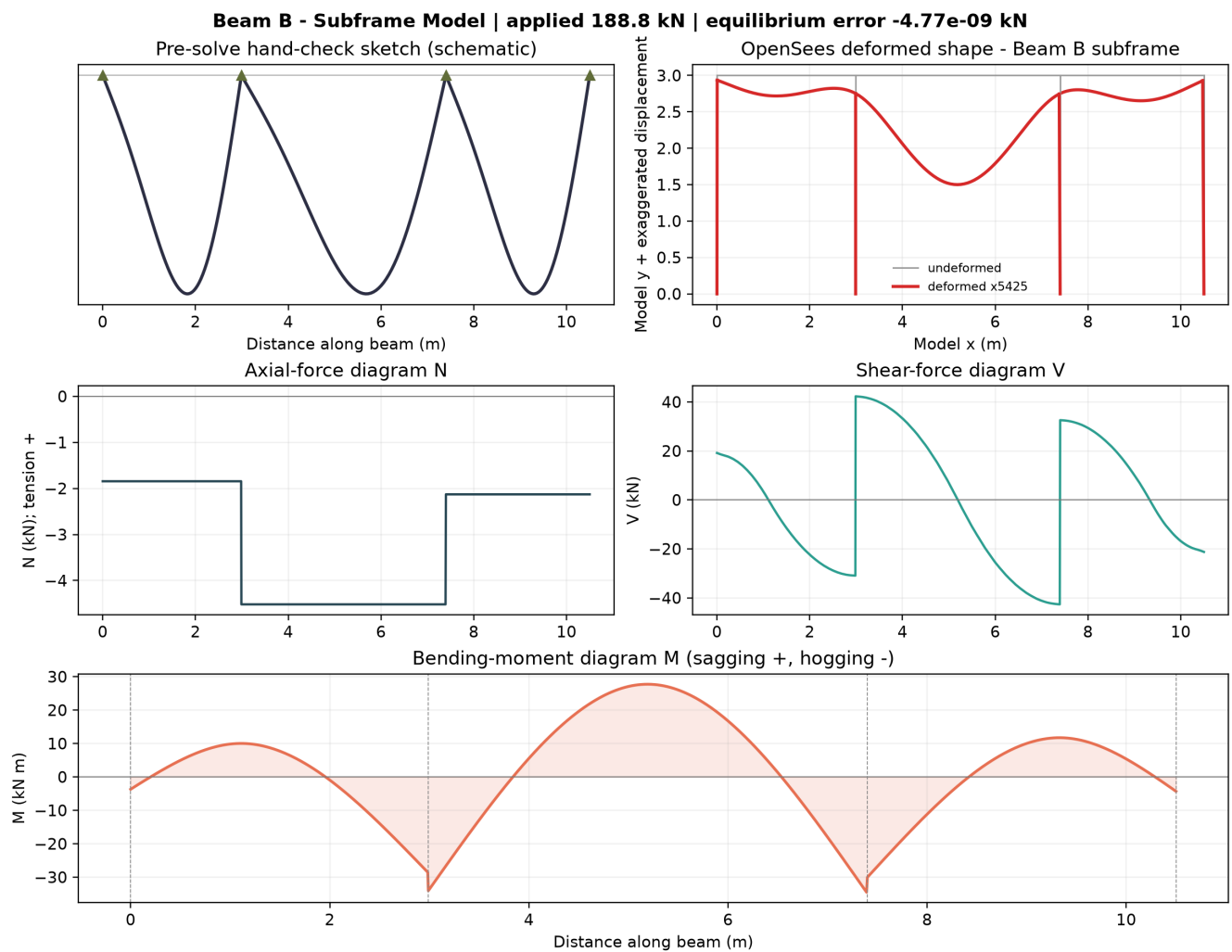


Figure. Beam B, subframe model: pre-solve sketch, deformed shape and N/V/M diagrams.

6. Numerical comparison

Beam	Model	Load	Sag +	Hog -	Max M	Down mm	qL2/8	Check
A	pure	130.245	31.504	-36.371	36.371	0.6002	47.687	PASS
A	subframe	130.245	23.786	-34.246	34.246	0.4377	47.687	PASS
B	pure	188.798	28.144	-34.678	34.678	0.2363	62.198	PASS
B	subframe	188.798	27.732	-34.523	34.523	0.2765	62.198	PASS

Loads are kN; moments are kN m. Every model closes equilibrium below $1e-8$ kN and has sagging-positive and hogging-negative regions. The subframe is the defensible beam-design model because it retains finite column stiffness; the standalone beam remains an independent bound.

7. HW1 axial cross-check

Strip	Grid	HW1 member	HW1 N	HW2 N	Diff %	Sign
A	G00	T1	-111.266	-24.952	-77.6	PASS
A	G01	C1	-32.699	-52.086	59.3	PASS
A	G02	C4	-42.360	-38.178	-9.9	PASS
A	G03	T10	-66.258	-15.029	-77.3	PASS
B	G01	C1	-32.699	-19.153	-41.4	PASS
B	G11	T5	-233.992	-73.199	-68.7	PASS
B	G21	C2	-36.226	-75.201	107.6	PASS
B	G31	C3	-27.321	-21.246	-22.2	PASS

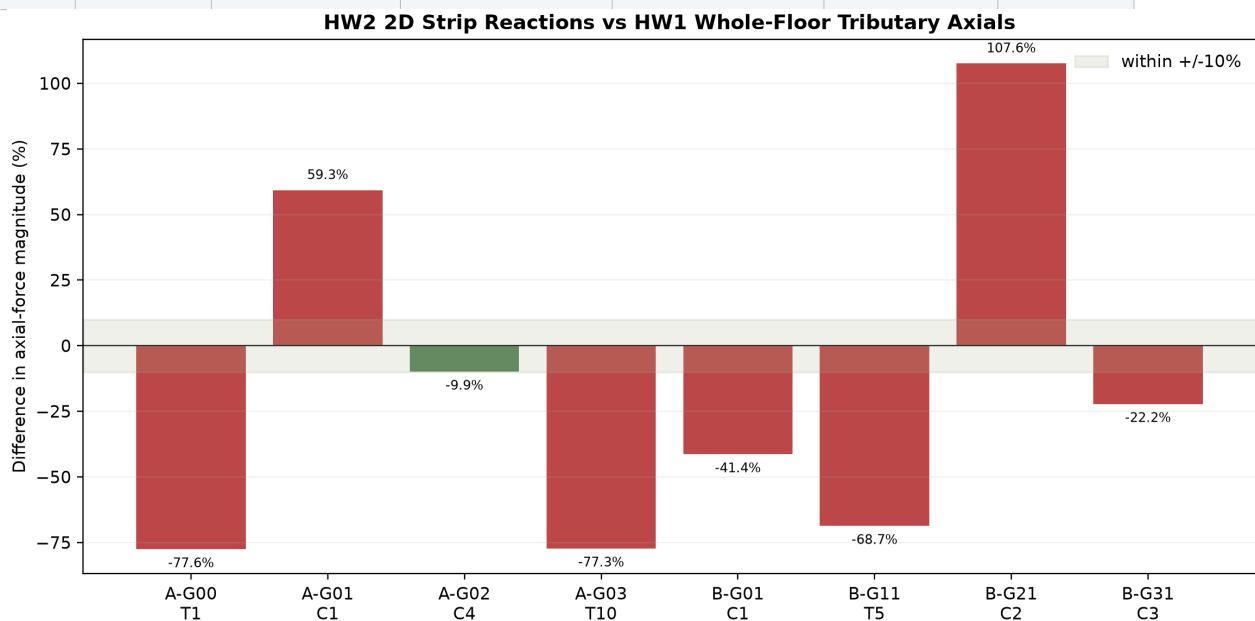


Figure. Magnitude difference between the 2D marked strips and HW1 whole-floor estimates.

Compression sign agrees at every mapped support, but the magnitude is not within a few percent. HW1 includes the whole-floor nearest-support load and aggregated walls; each HW2 result contains only one marked strip and omits perpendicular beam paths. No load was tuned. The subframe is used for the beam design question; exact cross-model reconciliation requires a 3D grid or common nodal aggregation.

8. Agent workflow and reproducibility

The word briefings were frozen before code generation and are embedded in the Markdown with the iteration record. The agent transcribed them into four OpenSees entry files plus an audited shared implementation. Gates were: classify the deck edges, freeze 45/60/30 loads, check units and signs, verify moment bounds, replace supports by columns, close equilibrium, compare HW1 axials, then verify the cumulative repository's HW2 package.

From repository root run: Set-Location HW2; then execute `analyze_hw2_beam_loads.py`, `run_hw2_models.py`, `summarize_hw2_models.py`, `build_hw2_report.py`, `render_hw2_report_pdf.py` and `verify_hw2_outputs.py` from scripts/.

Limitations

The plan is raster; beam size and uncracked stiffness are assumptions; and the subframes are 2D strips. Independently generated pre-solve hand-check schematics are placed beside every solver output.

Conclusion

All required beam census, panel support classification, 45/60/30 tributary areas, load- and moment-equivalent UDLs, four OpenSees cases, N/V/M and deformation diagrams, reactions, hand bounds, embedded prompt record, HW1 cross-check and reproducibility files are present in the HW2 package. The accepted HW1 package remains unchanged and separately reviewable in the same cumulative repository.